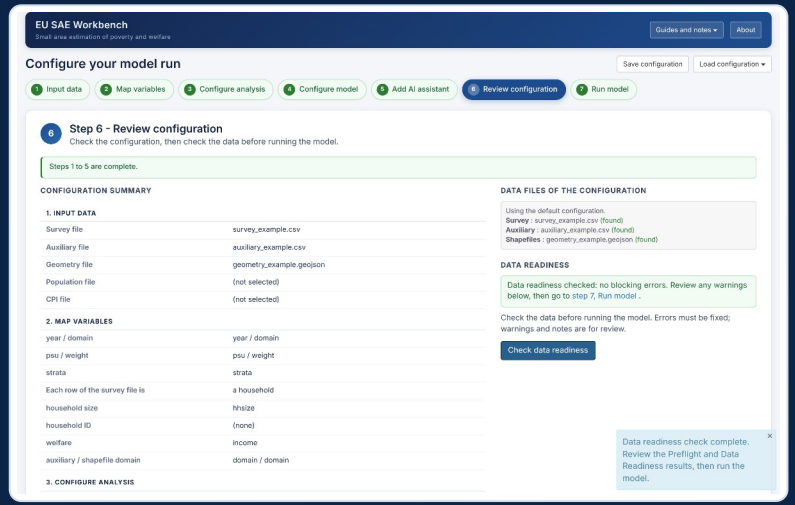
EU SAE WORKBENCH · GUIDE 2 OF 3

# Running the Workbench

The seven guided steps, from your data files to the report.



OVERVIEW

# Seven steps from data to results

Each step is one screen. You can move freely between them; Run model asks for a data readiness check first.

1. **Input data**
   Select the survey, auxiliary and map files
2. **Map variables**
   Say which column holds which variable
3. **Configure analysis**
   Two years and the indicator
4. **Configure model**
   UFH, MFH and benchmarking
5. **Add AI assistant**
   Optional report text
6. **Review configuration**
   Summary and data readiness check
7. **Run model**
   Progress, then the report

Front page: New model run starts at Step 1 with default settings. Load saved configuration fills every step and opens Step 6.

*The classic dashboard (Start_Dashboard) has the same settings on one page and runs the same analysis.*

# Three files are required

They can be in any folder. Use the same domain codes in all three.

## Survey
Household or person records

- year, domain code, PSU, survey weight and welfare
- household size, when each row is a household
- welfare per person (per capita or equivalised), not the household total
- optional: strata, household ID, poverty line, benchmark group

## Auxiliary
Area covariates

- one row per domain and year
- domain code, year and candidate covariates
- covers both analysis years
- may include areas without survey data

## Geometry
Area boundaries

- one polygon per domain
- a domain code matching the other two files
- shapefile: one .zip with the .shp, .shx and .dbf

Formats: R (.rds, .RData), CSV and other text files, Stata, SPSS, SAS, Parquet/Feather and Excel. Geometry: .rds, GeoPackage, GeoJSON, KML/GML or a zipped shapefile.

# Optional files

Add them only when the analysis needs them.

### Consumer price index (Step 1)

For mean welfare in constant prices. One row per year: a year column and the index. The values can also be typed in at Step 3.

### Domain population sizes (Step 1)

For benchmarking: one row per domain-year (long) or domain (wide). Duplicate keys are rejected, even for equal totals. Without a file, use survey totals.

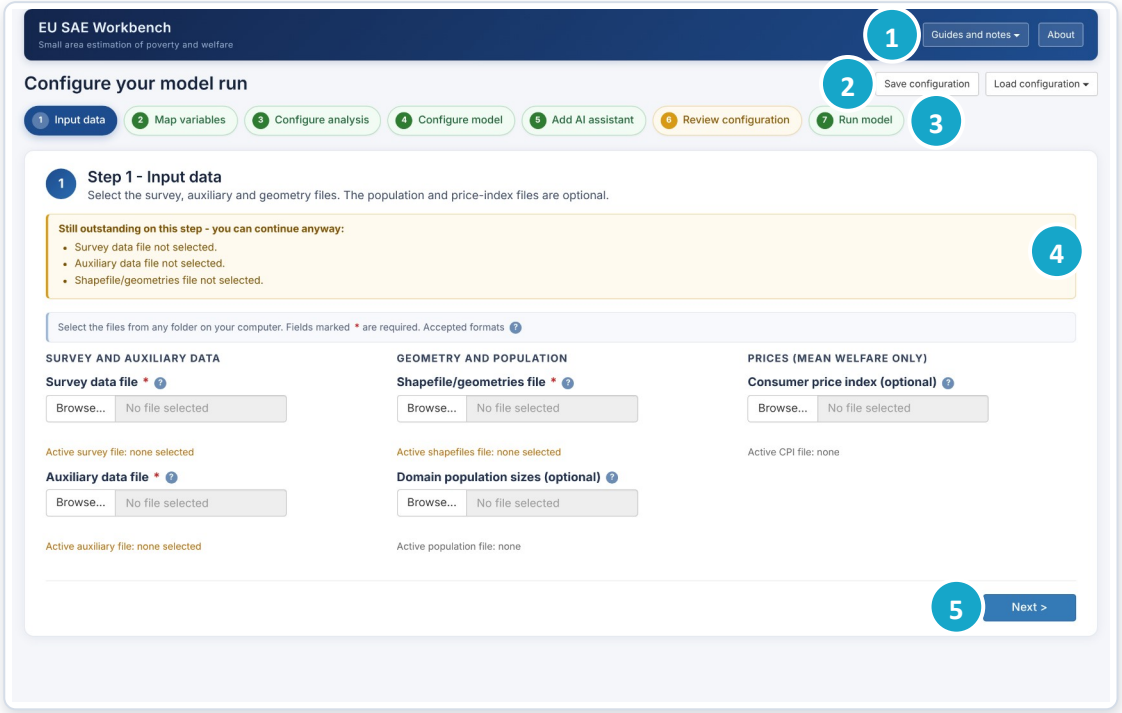
### Benchmark targets (Step 4)

External targets: one row per group-year (long) or group (wide). Duplicate keys are rejected, even for equal values. Without a file, use survey Direct estimates.

Named matrix columns match years. Unnamed columns require exactly the selected years, in order, with a warning.

# The screen at a glance

Every step has the same layout. Step 1 is shown before any file is selected.




1. **Guides and notes, About**
   Open the guides from inside the Workbench.

2. **Save and load configuration**
   Keep all settings for later or for another run.

3. **Step buttons**
   Green: done. Amber: missing items. Blue: current. Click to jump.

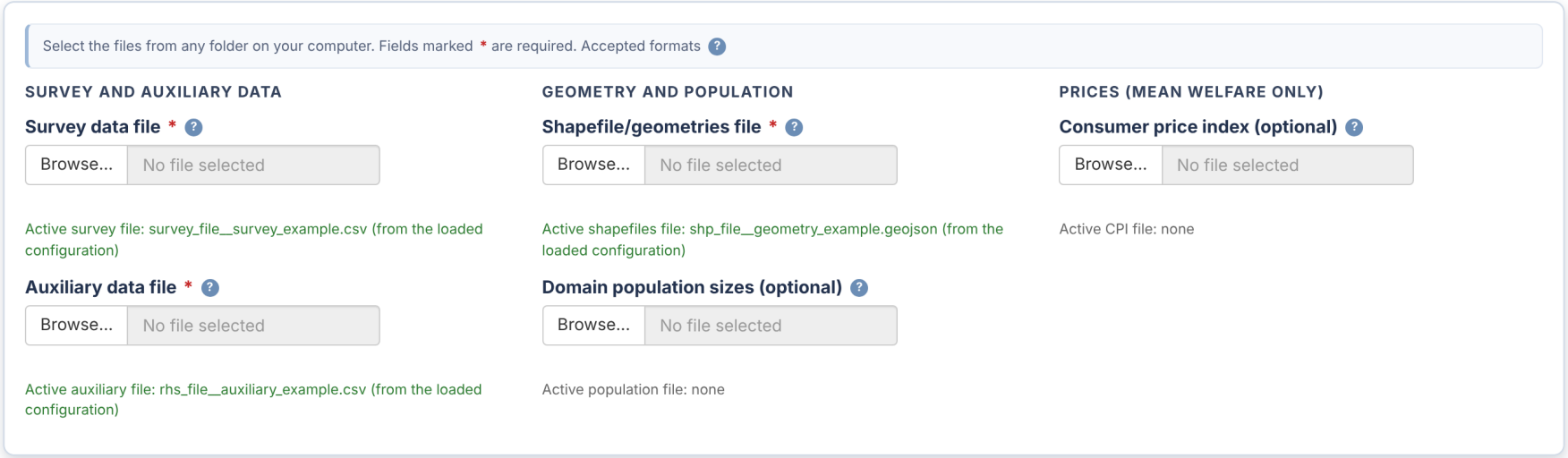
4. **What is still missing**
   Listed on each step. You can continue anyway.

5. **Back and Next**
   Move through the steps in order.

STEP 1

# Input data

Browse to each file. Required fields carry a red asterisk.




**Required**
Survey, auxiliary and geometry files.

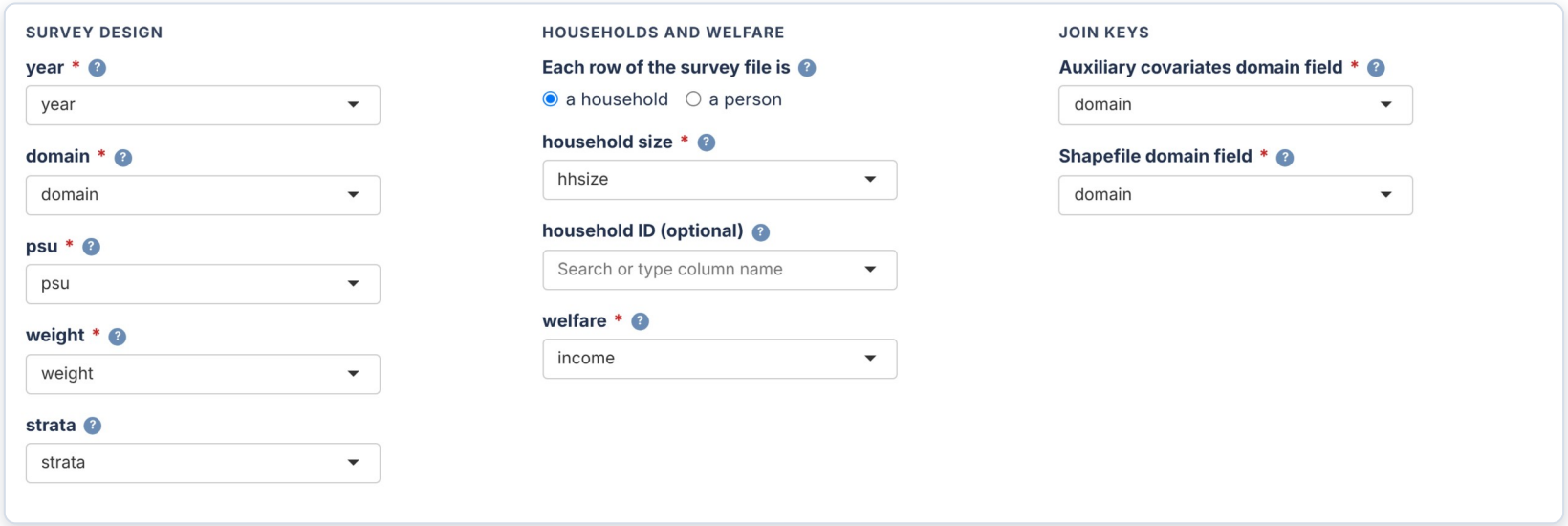
**Optional**
Domain population sizes; a consumer price index for mean welfare.

**After loading a configuration**
Its files appear under each Browse button. Browse only to change one.

STEP 2

# Map variables

Choose the column that holds each variable. The lists show the columns of your files.




**Survey design**
Year, domain, PSU and weight; strata if the design has them.

**Households and welfare**
What one row is, household size, and welfare per person.

**A field in red**
Names a column that is not in the selected file.

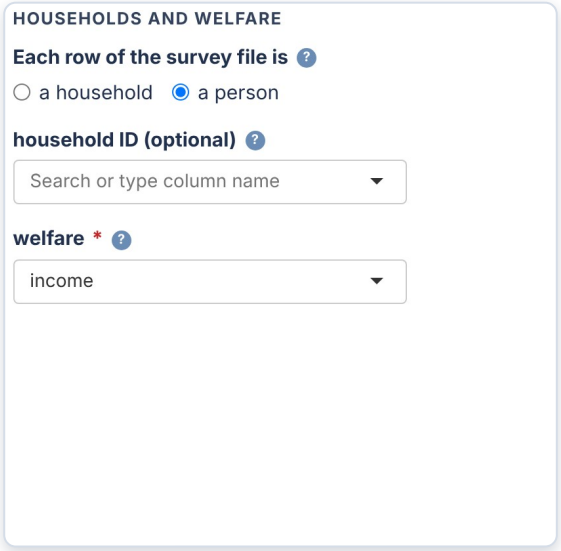
STEP 2

# Households or persons?

Tell the Workbench what one row of the survey file is. This decides how people are counted.




*A household: household size is required*




*A person: household size is not used*

**A household:** weight × household size counts every member.

**A person:** the weight is used as it is.

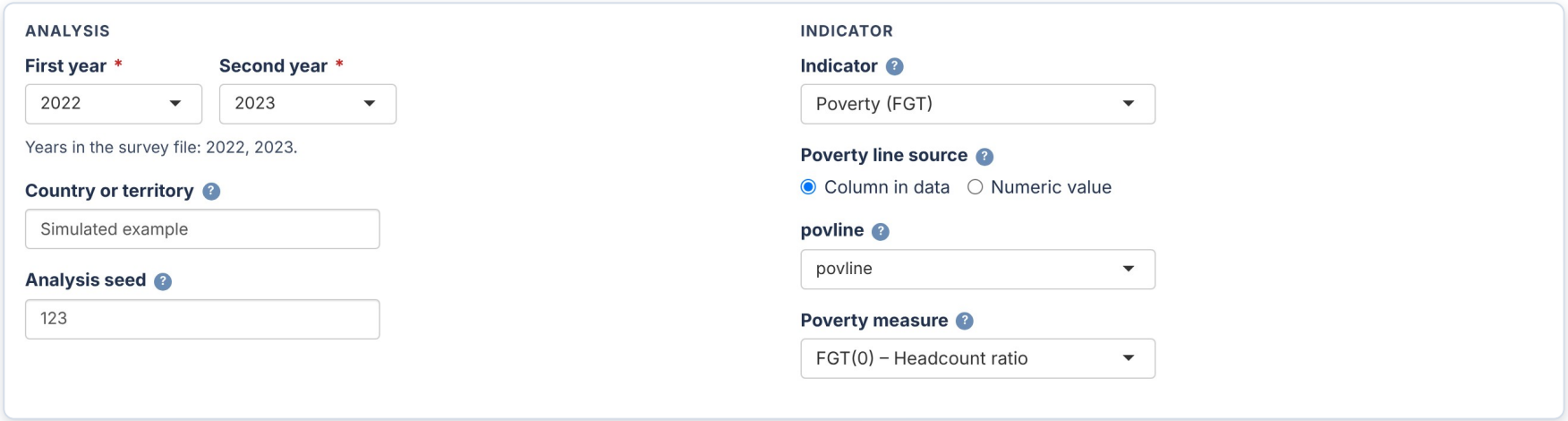
**Household ID (optional):** lets the readiness check confirm the choice.

A person file read as households counts each person once per household member.

STEP 3

# Configure analysis

Pick the two years to compare, then the indicator.




**Two years**
Chosen from the years found in the survey file.

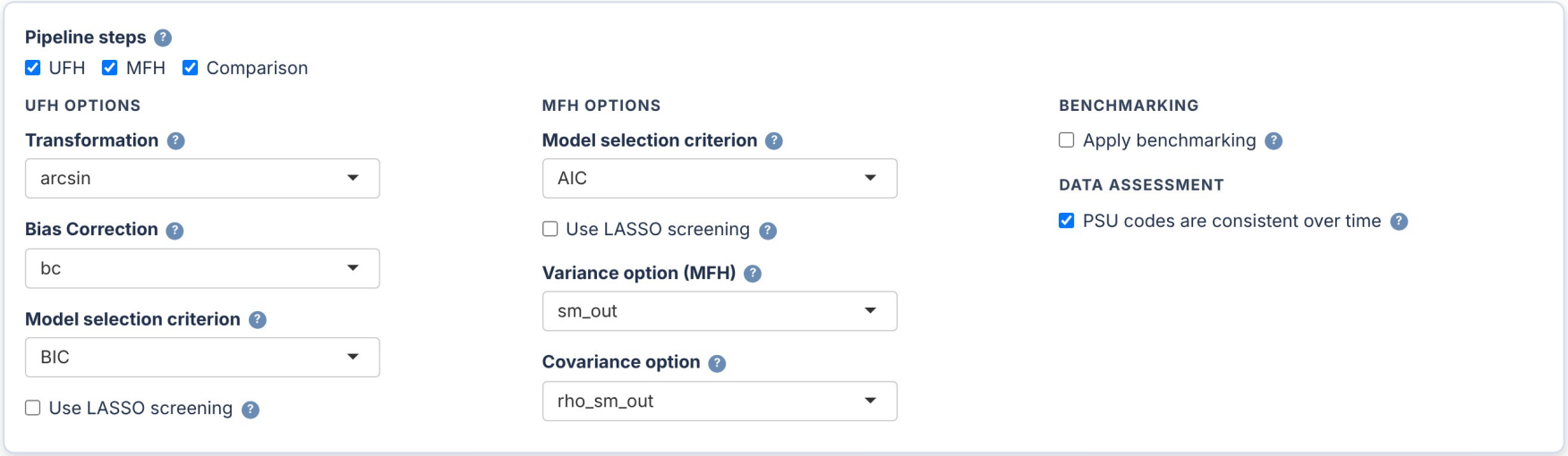
**Analysis seed**
The same seed, data, settings and package versions give the same results.

**Indicator**
A poverty measure (FGT) or mean welfare. Guide 3 covers both.

STEP 4

# Configure model

Current defaults below. The screenshot retains historical w5o settings.




**Pipeline steps**
UFH: one model per year. MFH: both years jointly. Comparison: both side by side.

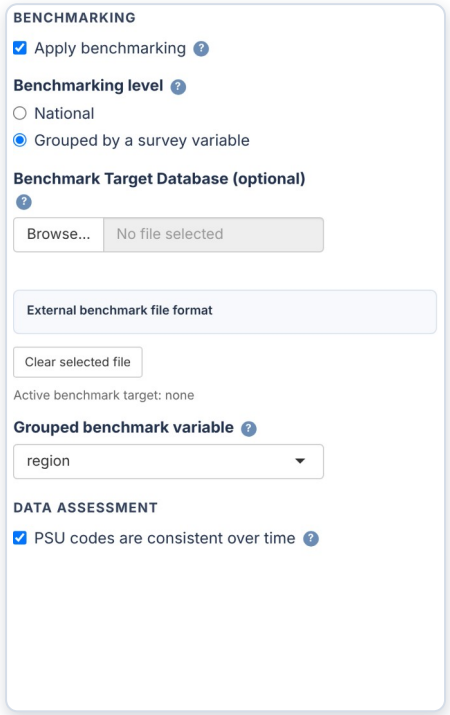
**Defaults**
MFH2, sm_all / rho_sm_all, 500 replicates. Explicit saved choices are retained.

**PSU codes consistent**
Under Data assessment: tick only if codes mean the same units in both years.

STEP 4

# Benchmarking (optional)

Makes the population-weighted average of the domain estimates equal the national or group figure.




**National:** one constraint for the whole country.

**Grouped by a survey variable:** one constraint per group, for example region or NUTS2.

**Each domain must belong to one group.** If a domain has two group codes, the readiness check names it, its codes, years and rows.

**Target file (optional):** same indicator and prices as the survey. Without it, the survey's direct estimates are the targets.

STEP 5

# Add AI assistant (optional)

Adds plain-language interpretation to the report. Off by default.




**Consent first**
Tick Enable and the consent box, then paste a Claude (sk-ant-…) or OpenAI (sk-…) key.

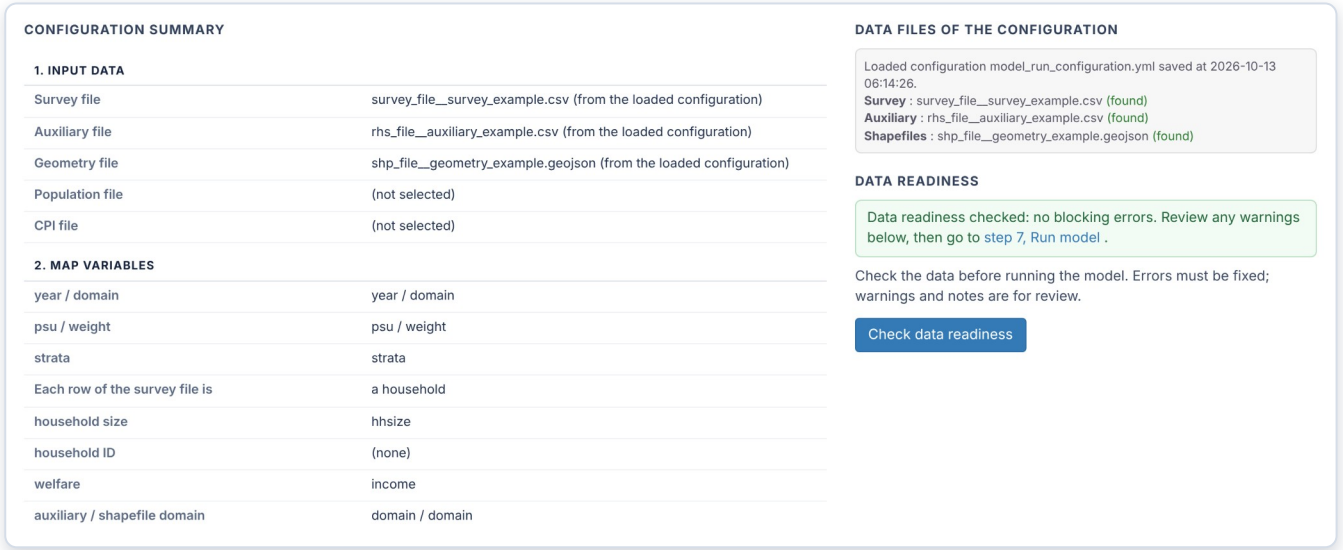
**What is sent**
Aggregate results only: no microdata, no area names or codes.

**Key, language, Comparison**
The key is not saved. 24 EU languages and Arabic. Runs with the Comparison step.

STEP 6

# Review configuration

Read the summary, then press Check data readiness.



CONFIGURATION SUMMARY

| 1. INPUT DATA | |
|---|---|
| Survey file | survey_file__survey_example.csv (from the loaded configuration) |
| Auxiliary file | rhs_file__auxiliary_example.csv (from the loaded configuration) |
| Geometry file | shp_file__geometry_example.geojson (from the loaded configuration) |
| Population file | (not selected) |
| CPI file | (not selected) |

| 2. MAP VARIABLES | |
|---|---|
| year / domain | year / domain |
| psu / weight | psu / weight |
| strata | strata |
| Each row of the survey file is | a household |
| household size | hhsize |
| household ID | (none) |
| welfare | income |
| auxiliary / shapefile domain | domain / domain |

DATA FILES OF THE CONFIGURATION

Loaded configuration model_run_configuration.yml saved at 2026-10-13 06:14:26.
**Survey** : survey_file__survey_example.csv (found)
**Auxiliary** : rhs_file__auxiliary_example.csv (found)
**Shapefiles** : shp_file__geometry_example.geojson (found)

DATA READINESS

Data readiness checked: no blocking errors. Review any warnings below, then go to step 7, Run model .

Check the data before running the model. Errors must be fixed; warnings and notes are for review.

Check data readiness

**Summary**
Every setting on one page, and where each data file comes from.

**Check data readiness**
Reads the data and runs the checks. The results appear below the summary.

**Changed something?**
Check again: the app asks for it after changes to data, mapping, years, indicator or prices.

STEP 6

# Reading the data readiness results

ERROR lines must be fixed before the run. WARNING lines and notes are for review.

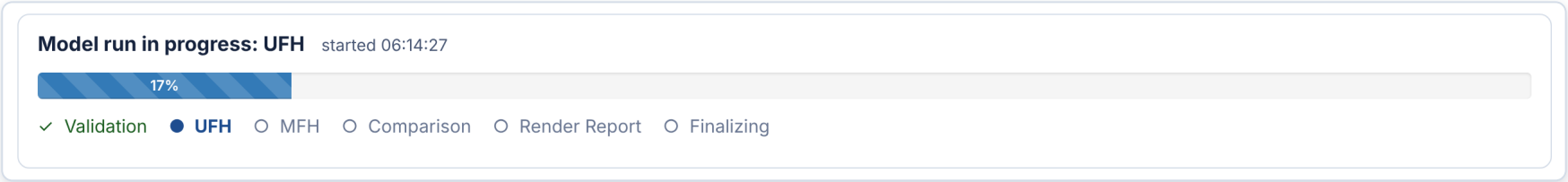
| Test | What it checks |
|---|---|
| **0a, 0b** | A year column in both files; both analysis years present in the survey and auxiliary files |
| **0c** | Weights and household sizes give valid population weights |
| **0d, 0g** | Grouped benchmarking: the group variable exists; each domain belongs to one group |
| **0e** | Mean welfare in logs: zero or negative welfare, and positive area means |
| **0f** | Households or persons: the weighted population total and, with a household ID, the row type |
| **0h** | Consumed population/target inputs: unique keys, domain/group coverage and selected years |
| **1** | Covariates: means and correlation with the indicator |
| **2, 3** | Domain coverage: MFH requires a Direct estimate for every domain in both years |
| **4** | The national poverty rate or mean welfare in each year |

MFH missing domain-years and invalid required inputs block the run before output cleanup. Unused optional paths do not block it.

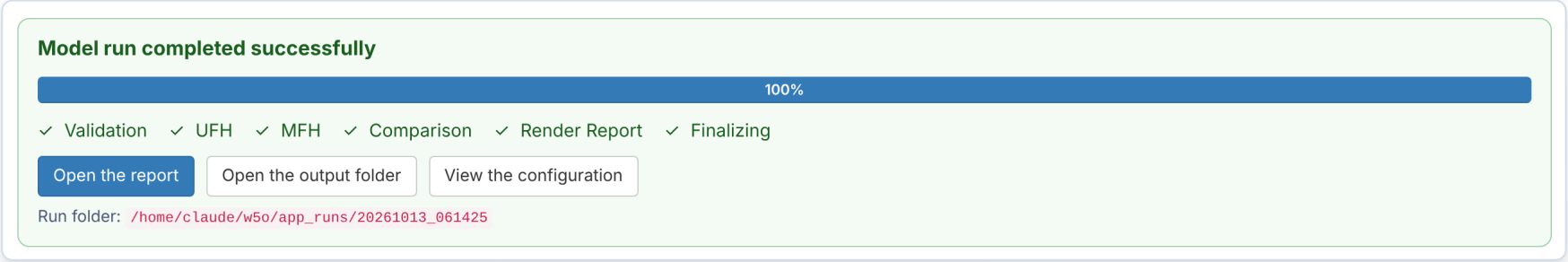
STEP 7

# Run model

The run continues in the background. The page shows each stage, then the result.




*While the model runs: the stage and its progress*




*When it is complete: open the report, the output folder or the configuration*

Stop ends a run. If a run fails, the panel shows the reason and a button to open the run folder.

# Where the results are

outputs/ holds the latest run. Every run is also kept in its own folder.

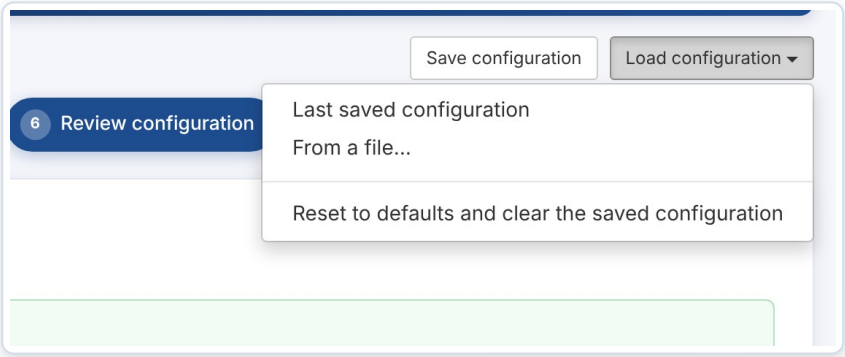
| File or folder | What it contains |
|---|---|
| **outputs/final_report.html, .docx** | The report: maps, tables, diagnostics and changes. The Word file is editable. |
| **outputs/data/** | Excel workbooks with Direct, UFH and MFH estimates, MSE, RMSE and CV, and the changes |
| **outputs/figures/** | Maps and charts as PNG files |
| **outputs/tables/** | CSV tables: readiness, fit checks, bootstrap diagnostics and ufh_variance_audit.csv |
| **app_runs/<date_time>_<label>/** | A copy of the run's outputs, its log and model_run_configuration.yml |

A validated run replaces outputs/. To edit the Word report, work on a copy, or use the copy under app_runs/.

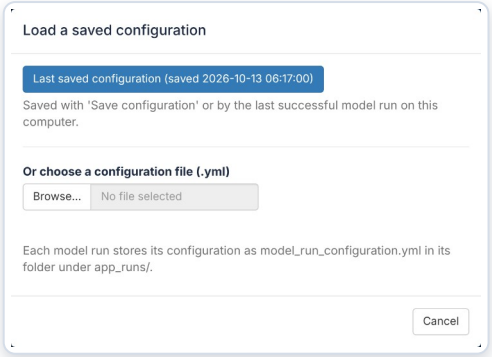
Direct MSE is raw survey-design variance. ufh_variance_audit.csv separately records the variance used by the model, its scale and policy.

# Save, load and repeat a run

A configuration holds every setting and where the data files are. Reuse it instead of starting again.




*Load configuration menu, on every step*




*Load saved configuration, on the front page*

**Save configuration:** keeps the settings and copies of the data files in app_runs/_last_setup_files.

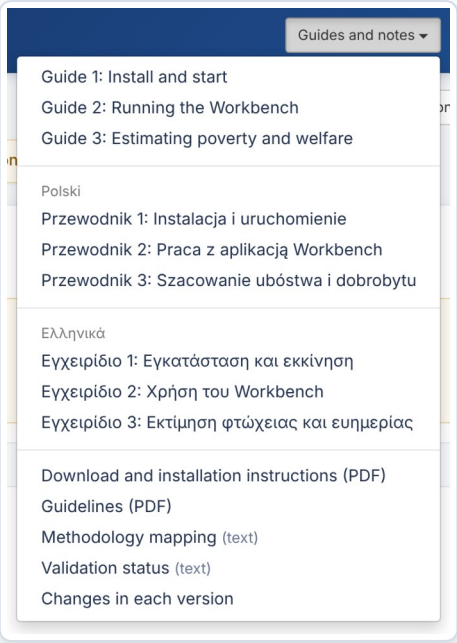
**Load:** the last saved configuration, or a file such as a run's model_run_configuration.yml. The review step opens.

**Changed data file?** Loading warns if a file differs from the one the configuration was saved with.

**Reset to defaults** deletes the saved file copies; older run configurations then need their files browsed again.

# Help inside the Workbench

Three places to look before asking.




*Guides and notes menu*

**?** **Question marks**

Hover over the ? next to a field for what it means and which values it accepts.

**G** **Guides and notes**

The three guides in English, Polish and Greek, the guidelines note, the methodology mapping, the validation status and the changes in each version.

**i** **About**

The version you are using.

Problem reports: github.com/Noboyoshida227/eu-sae-application/issues. Attach the run log, never the data.

CHECKLIST

# Every run, in short

Before you report results, check more than the map.

| Step | Check |
|---|---|
| **1 Input data** | The three required files; CPI or population file if needed |
| **2 Map variables** | No red fields; household or person rows; household size |
| **3 Configure analysis** | Two years from the survey; indicator and poverty line or prices |
| **4 Configure model** | Defaults unless you have a reason; PSU answer; benchmarking |
| **5 Add AI assistant** | Off, or consent and key |
| **6 Review configuration** | Summary read; readiness without ERROR |
| **7 Run model** | Report opened; diagnostics, bootstrap results and changes reviewed |

Guide 3 shows the settings and the reading of results for poverty, anchored poverty and mean welfare.